

Equation of a plane

Key Points

A plane is a point a and two in-planar direction vectors d_1, d_2

$$r = a + \lambda d_1 + \mu d_2$$

Directions d_1, d_2 can be found from two points in the plane. The normal to the plane is $n = d_1 \times d_2$ and so

$$(r - a) \cdot n = 0 \text{ for constant } d$$

$$\text{So for } n = (a, b, c)$$

$$ax + by + cz = d$$

With d found by substitution of a point on the plane

Basic Skills

Q1. Find a normal of the two vectors $\begin{pmatrix} 3 \\ -1 \\ 2 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}$

Q2. The plane Π contains the points $(1,3,5), (2,3,8), (0,1,10)$

- Determine two in-planar vectors
- Determine the normal to the plane by taking the vector product of your two in-planar vectors
- Hence write a vector equation for the plane in the form $(r - a) \cdot n = 0$
- Hence write the cartesian equation for the plane
- Decide if the point $(3,3,11)$ is in the plane
- Decide if the origin is contained in the plane

Competency Skills

Q3. A plane is defined by a point $(0,3,-2)$ and has a normal of $(1,0,-3)$ find the cartesian equation

Q4. A plane is defined the cartesian equation $2x + 5y + 3z = 60$

- Find a unit normal to the plane
- Write the vector equation of the plane

Q5. Deduce if the origin is in the plane containing the points $(4,5,1), (2,3,0)$ and $(-1,-1,-1)$

Q6. What condition makes a plane parallel to another

Q7. Plane Π_1 is given by $2x - 4y = 15 - z$, find the parallel plane Π_2 in cartesian form that contains the point $(5,2,10)$

Advanced Skills

Q8. Show that the constant of the cartesian equation can be found by $d = a \cdot n$

Q9. Find the cartesian equation of a plane containing the point $(1,0,1)$ and the line $\frac{x-1}{2} = \frac{y+3}{2} = z - 1$

Q10. Explain why the points $(-5,1,2), (5,2,4)$ and $(25,4,8)$ do not form a plane

Q11. Find the plane containing both the lines $r_1 = \begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 1 \\ -3 \end{pmatrix}$ and $r_2 = \mu \begin{pmatrix} 3 \\ 2 \\ 0 \end{pmatrix}$

Q12. Find the point at which the plane $2x + 5y + z = 10$ and the line $y = x + 4 = 2z$ intersect

Extension

Q13. Consider if two planes always intersect

Hint: Normals to planes

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